

Discovering of Governing Equations DMD and SinDy for Nonlinear Physica System The Kinetics Sculpture

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Abstract—The Swinging Sticks, a double-pendulum kinetic sculpture, represents a canonical example of nonlinear dynamics with strong sensitivity to initial conditions and complex energy transfer between coupled arms. Traditional physics-based models derived from Lagrangian mechanics capture their qualitative behavior but remain analytically intractable and highly sensitive to noise in experimental data. Following publishing of dataset of the kinectic sytsem

Index Terms—Chaotic System, Dynamical System, Nonlinear System, Attractor, Hough Transform, Machine Learning

I. INTRODUCTION AND MOTIVATION

The Swinging Sticks sculpture is an engineered double pendulum system designed to resemble a perpetual motion machine. From the popular scene in Iron Man 2 movie, like the double pendulum, its behavior has long fascinated physicists and engineers, inspiring the exploration of even more complex systems in nature: three body problems and chaos in physical systems. Its motion is sustained by sensors and magnets, yet the underlying physics is that of two coupled pendulums with very low damping [1]. Multidegree-of-freedom pendulums, from the classic double pendulum up to higher arms-systems form the traditional examples of nonlinear dynamical systems. These systems exhibit a wide range of behavior, from simple harmonic oscillations at low energy, unpredictable oscillations or aperiodic trajectories to fully chaotic motion at higher energy levels [2]. In particular, increasingly minute variations in initial angles lead to completely different trajectories: the swinging kinetic system, like the double-pendulum's motion, is famously chaotic, with strong sensitivity to initial conditions, posing a huge challenge to its predictability. There is also the importance of accounting for energy dissipation and external forces to avoid unrealistic predictions, an additional layer of complexity to the dynamics of the problem of this system. Adding more links increases complexity; multi-arm pendulums develop intricate phase-space structures that mediate transitions between different oscillation modes. These similar features in the swinging kinetic sticks make it challenging as benchmark for nonlinear dynamics and control studies.

Results from machine learning research indicates the community lacks benchmarks for chaotic systems from real chaotic physical systems, which can be leveraged to test the reliability of predictive models. An accurate and adaptive model is required to fully understand the dynamics of the Swinging Kinetic system, as, unlike a simple double-pendulum system, the equations that govern the dynamics of its behavior are unknown. In this paper, an ablation study on time series



Fig. 1. The Swinging Kinetic Sticks

prediction is performed using the top five best-performing time series predictor models from the literature on a novel dataset from the Swinging Kinetic Sculpture. Post identification of best model predictor, Dynamic Mode Decomposition (DMD) and Sparse Identification of Nonlinear Dynamics (SINDy) are proposed to be leveraged for the development of data-driven models for discovering the governing dynamics or equations of the Kinetic Swinging Sculpture.

A. Related Works

Classical approaches for discovering the dynamics of similar coupled systems is the Lagrangian or Hamiltonian formalisms. By choosing the limb angles as generated coordinates, one can derive the coupled nonlinear equations of motion from first principles [3]. As such, the double pendulum Lagrangian is given by the kinetic minus the potential energy in terms of the two angles, leading to coupled second-order ODEs for the angles. These equations capture energy conservation and the full nonlinear coupling, but admit closed-form solutions only in special limits; small-angle approximations. In practice, analytical perturbation or numerical integration of these ODEs is used to understand mode shapes and energy transfer in the pendulum system, demonstrating effectiveness, but often being computationally intensive to tune for accuracy. Pendulum systems are known for their chaotic behavior as they exhibit bifurcation. In such models, the pendulum arms are abstracted into coupled oscillators, while torques are applied as input to control angular positions and velocities. The model effectively captures the effect of total kinetic energy on the phase space.

Koopman/DMD-based techniques take a different tack, embedding the nonlinear pendulum flow into a high-dimensional linear system. Applied to Lorenz and Rossler chaotic sys-

tems, and even a double pendulum, HAVOK can separate the dynamics into a dominant linear part with intermittent forcing [4]. In practice, these often achieve strong predictive accuracy and generalization. For instance, Berrueta et al. (2020) demonstrated that a polynomial EDMD (extended DMD) model of a simulated double pendulum had the lowest average error across state-space among several regressors [5]. However, such models yield only a linear surrogate of the dynamics (not explicit ODEs). By contrast, deep recurrent networks (RNN/LSTMs) can directly learn the time series with remarkable short-term accuracy. In one study, an LSTM model predicted the next 1.5s of a chaotic double-pendulum trajectory within a tight error bound, whereas a best-fit analytic model diverged in just 0.12s [6]. The drawback is that these neural models are opaque “black boxes” that do not produce human-readable equations [7].

For prediction, with the ideal embedding dimension determined, reconstruction and forecasting of a chosen model can be performed. By comparing predictions from real data to those from surrogate data, using Takens’s embedding in its application of dynamic reconstruction and nonlinear prediction techniques to laboratory-generated geophysical data, Smith refines earlier methods of detecting low-dimensional chaos [8], using this method in the challenging task of distinguishing between deterministic and stochastic behaviors in low-dimensional dynamic systems.

B. Contribution and Novelty

This section outlines the novel contributions of this work, including the data acquisition process for the continuous study of the seemingly non-terminating motion of the Swinging Kinetic Stick sculpture and the first ablation study on the prediction of timeseries from its timeseries. Below are other major contributions of this study:

- Published the first dataset of the motion of the Swinging Kinetic Sculpture System
- Introduced an approach for studying the overall evolution of the system until termination
- Proposed method for discovering the dynamics and governing equation of the system

C. Objective

In this paper, the underlying behavior of the chaotic motion of swinging sticks is studied by predicting its next state from its timeseries and ultimately determining the best nonlinear model predictor ideal for timeseries prediction and forecasting of state of its dynamics. There is no known baseline best prediction model for reference on the predictability of swinging sticks timeseries. In the work, we seek to identify and establish this baseline for extended studies on this system.

II. THE SWINGING KINETIC SYSTEM CHAOTIC DATASET

A. Mechanical System

The swinging rods, also known as kinetic motion toys or kinetic sculptures, have captivated the imagination of both scientists and enthusiasts for decades. These mesmerizing contrap-

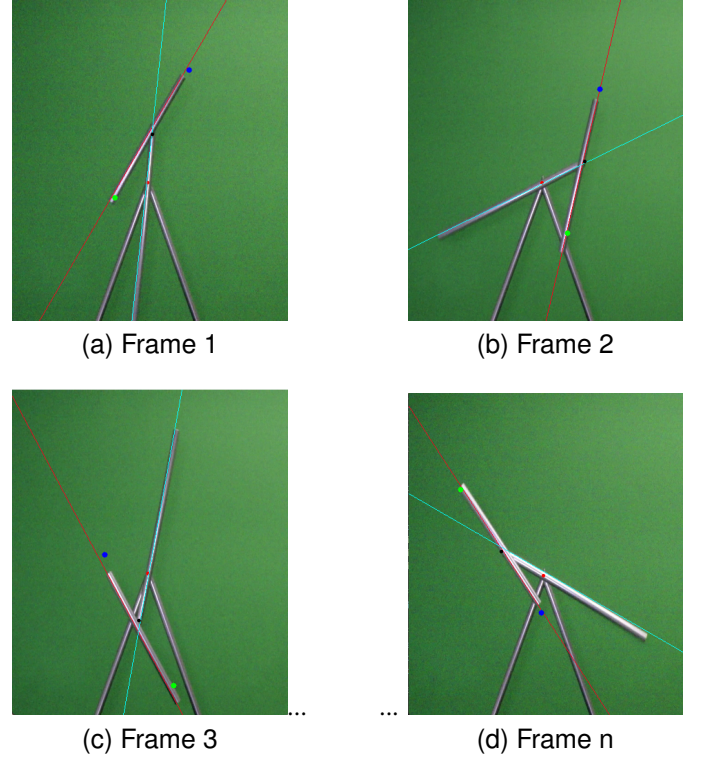


Fig. 2. A sequence of frames showing red line and green line tracking the motion of the short and long stick respectively. The red point is the center of the system. The blue and light green spots around the short stick serve as a guide for determining the short end of the short stick

tions consist of two (long and short) metallic rods(pendulums) connected by hinges that allow rotation. The system has a triangular support made up of stationary standing legs or edges with their meeting point serving as a center of rotation of the long rod. The system is driven by an electromagnetic motor, driven by a magnetic field that initiates and sustains oscillations. The swinging rods have a powerful neodymium magnet in them that enables them to exhibit complex nonlinear dynamics due to their coupled motion and sensitivity to initial conditions. When set in motion, they exhibit complex rhythmic patterns, seemingly defying gravity as they gracefully swing and rotate. The dynamics governing the motion of these rods are not only fascinating from an aesthetic standpoint but also hold profound implications like every double pendulum system in physics, engineering, and mathematics with rich dynamic behavior. Exploring the underlying principles behind these dynamics not only enriches our understanding of mechanical systems, but also provides insights into broader scientific and technological domains.

B. Data Acquisition Process

The process involved using a high-quality camera and video recording the motion of the sticks with a resolution of 1920 x 1080 at 60 fps. The exposure of the lens was set high to allow light to be captured; hence the light setting was made normal and augment with an extra DC power LED to illuminate

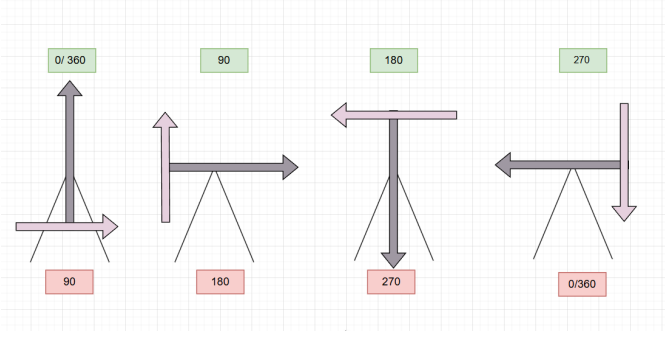


Fig. 3. The graph shows the orientation configuration used for tracking the evolution of motion of the short stick (angle in red) and long stick (angle in green) of the system. The longer end and the shorter ends of the long and short sticks, respectively served as their pointers

TABLE I
CAMERA SETTINGS.

Camera settings	
Focal length [mm]	50
Frame exposure time [μ s]	90
Capture frame-rate [FPS]	60
Image resolution [px]	1920 $\times$ 1080

the room. To allow the easy extraction of the sticks position and to avoid shadow interference, a green screen background was used. For each of the measurements explained below, the system's initial condition on start involves displacement of the Swing sticks or rod, long and short, at 0° and 90° , respectively. The initial configurations defined from the angles of the sticks determined are shown in figure.

C. Long and Short Arm Positions Extraction

The raw video was run through an OpenCV based image processing program that utilizes details like center of the frame, long stick perpendicular distance from the center to extract the positions and ultimately the angles of rotation of both long and short sticks. This is realized using the Hough transform detection of long and short rod's orientation via line detection. The Hough transform function outputs all straight lines detected in each frame in ρ, θ pairs, where ρ is the perpendicular distance the lines make from the top left corner of the frame and θ is the angle the perpendicular to the lines makes with the horizontal frame. The ρ, θ values were further used to determine the endpoints of the lines and their actual orientation using an algorithm in python which writes the angles of rotation of the long and short sticks into file.

The timeseries data gathered from this system serves as data from a real-world system that we ultimately will test our improved forecasting models against. For this physical system, our interest in it as a chaotic system is the number of times the long stick crosses the top and also the equations of its dynamics. The preliminary results are presented in the results section however the algorithm is detailed below.



Fig. 4. Shows the hardware setup for the data acquisition process; with lamp stand, camera, the kinetic swinging sculpture and a green background to enable improved image processing

III. METHODOLOGY

A. Experimental System and Setup

The first attempt at gathering data from such a chaotic physical system done in this section. The process resulted in two sets of times series data representing the rotation of the long and short sticks of the system. The experimental setup involved the following:

- *The Kinetic Swing Sticks*: A commercial kinetic energy system with two freely rotating sticks.
- *Measurement Devices*: A high-resolution camera (60 fps or higher) is used to capture the motion of the system. The camera is positioned to record the side view ensuring that both the angular displacement and velocity of the rods are visible.
- *Data Acquisition System*: A Lenovo laptop is connected to the camera to collect data of the motion of the sticks in real. A text file is also used in collecting the data produced from the detection of rod motion detection algorithm.

B. Image Processing Algorithm

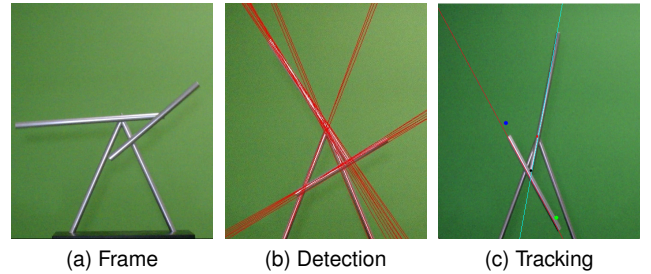


Fig. 5. These frames show the major stages of the angle position acquisition algorithm which processes angle positions of short and long sticks frame by frame. For each frame(a), hough transform detects lines which are grouped and ultimately tracked frame by frame

Algorithm 1 Sticks Angles Data Acquisition per Frame

Require: Recorded video (1920×1080)

Ensure: *long_stick_angle*, *short_stick_angle*

- 1: Resize frame to 640×480; set center [320, 340]
- 2: Apply Hough transform to obtain (ρ, θ)
- 3: Filter stands: left $\theta \in [18^\circ, 24^\circ]$, right $\theta \in [155^\circ, 163^\circ]$
- 4: Remove stand lines; retain stick lines
- 5: Group and average stick angles
- 6: Convert (ρ, θ) to Cartesian form
- 7: **for** angle transitions in $[175^\circ, 185^\circ]$ **do**
- 8: Add 180° for continuity
- 9: **end for**
- 10: Compute line equations: $y = m_1x + b_1$, $y = m_2x + b_2$
- 11: Compute intersection:

$$x_{poi} = \frac{b_2 - b_1}{m_1 - m_2}, \quad y_{poi} = m_1x_{poi} + b_1$$

- 12: **return** *frame_number*, *time*, *long_stick_angle*, *short_stick_angle*, *ott*

An elaborate version of the algorithm is provided in the appendix ??.

C. Data Cleaning and Smoothing

Considering the rotation of the angles is sinusoidal in nature a quadratic polynomial smoothing technique would be ideal for the smoothing process. This is verified by comparing the quadratic polynomial fit against linear fit for the pure sine wave, noisy sine wave and sticks long angle. For the smoothing process, a sliding window approach is adopted with a window size of 11 applied to a 60-frames-per-second time series consisting of 60 angle measurements. The fit is performed on each set of 11 consecutive angles, after which the window is shifted by one step. This process is repeated until all angles are processed. This produces a distribution of fitted values for each angle, with the first and last angles being fitted only once. The results show that linear fits exhibit a higher residual range and perform overall worse compared to quadratic fits, which closely approximate the true values. The method was subsequently tested on actual stick data, with results shown in Figure 6 and Figure 7. Throughout these plots, the angles are also referred to as observed values. Gaussian noise with a standard deviation of 5 is added.

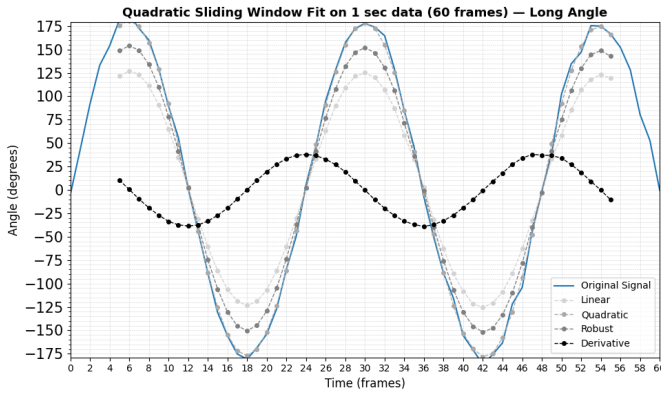


Fig. 6. The graphs illustrates the choice of robust sliding window quadratic smoothing technique for removing data fluctuation. Cleaning of data is performed on 60 frames of data from the long and short stick motion

D. Data Generation

The timeseries of the long and short sticks were generated from the system with an initial condition of long stick displacement angle of 90 degrees while the short stick remained at rest. A fifteen hours and twenty hours continuous data data acquisition process produced 3 million and 5 million data points respectively. The resulting time series used for this ablation study has $N = 3000$ points. Total series length of $N = 3000$ is consistent with the literature used for Ikeda [9], [10].

E. Embedding and Feature Construction

Supervised input-output pairs were constructed via Takens' delay embedding theorem [11]: with embedding dimension $d = 5$ and delay $\tau = 1$. All inputs and outputs were standardized to zero mean and unit variance.

$N_{\text{train}} = 2000$ consecutive samples were used for training (ordered in time; no shuffling to preserve dynamical structure), followed by $N_{\text{test}} = 1000$ for evaluation. This 2:1 ratio follows standard practice in chaotic benchmark studies.

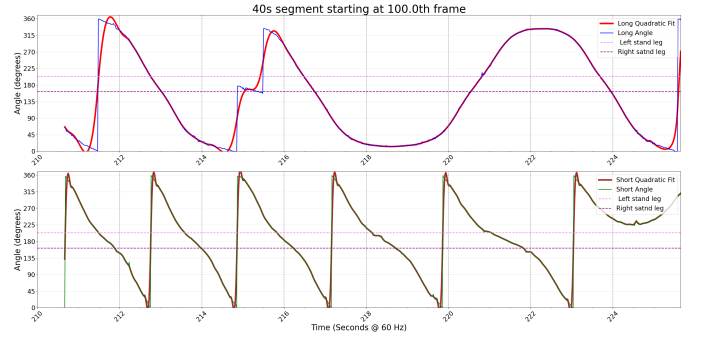


Fig. 7. The graph on the top shows a section of raw(blue) and cleaned(red) of long stick timeseries. The bottom graph shows the raw(brown) and cleaned (green) timeseries of the short stick. The dotted lines on both graphs show the position of the left and right stands of the system

Three complementary metrics are reported:

- **RMSE:** $\sqrt{\frac{1}{N} \sum_i (y_i - \hat{y}_i)^2}$
- **NRMSE:** $\sqrt{\frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}}$ (variance-normalised; model is useful if $\text{NRMSE} < 1$)
- **$R^2(\text{cod})$:** $1 - \text{NRMSE}^2$
- **MAE:** $\frac{1}{N} \sum_i |y_i - \hat{y}_i|$

IV. MECHANICAL MODEL AND COORDINATES

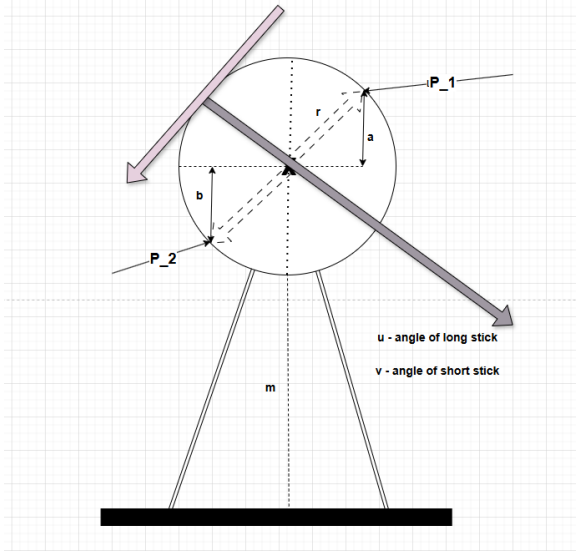


Fig. 8. Coordinates and Parameter of interest of the the Swinging Kinetic Sticks mechanical system. r is the axis of rotation of the long stick and e for nonstationary axis of rotation of the short stick

In this section, the process for turning simultaneous angular position and angular velocity timeseries of the two sticks(rods) into their corresponding Potential Energy $U(t)$, Kinetic Energy $V(t)$ and Total angular momentum $L(t)$.

The system dynamics of the system can be modeled first as two rigid uniform rods rotating about a single fixed axis. The long rod (angle θ_L) and the short rod (angle θ_S) each have one rotational degree of freedom; they are coupled only through the near-hub neodymium magnets. By simplifying the system and considering short stick as having same axis of rotation for the long stick(at point P_1) and short stick(reaching reached P_2), each rotating independently in one plane, magnetically coupled at the central axis of rotation O . Assume short stick has not coupled to long and determine it potential energy later the axis of rotation of the shorter stick can be later treated as non-stationary.

System coordinates: $q = (\theta_L, \theta_S)$.

The tracked angle is in degrees, zero pointing up, increasing clockwise or decreasing counter-clockwise. For mechanics we want radians and a right-handed convention. Define, per rod,

$$\varphi_i(t) = \varphi_i^0 - \frac{\pi}{180} \theta_i^{\text{clean}}(t), \quad i \in \{L, S\} \quad (1)$$

To determine the kinetic and potential energy of each of the sticks, for each stick in pure rotation motion, calculate the following:.

1) Center of Mass Position (CM):

$$y_{\text{cm},i}(t) = y_0 + d_i \cos \beta_i(t), \quad (2)$$

where $\beta_i(t) = \varphi_i^s(t) + \gamma_i$ is the angle of the CM position vector from the upward vertical, and $\gamma_i \in \{0, \pi\}$ accounts for the CM lying on the pointer side of O . y_0 is an arbitrary initial or starting point of CM and abstracting out of all dynamics.

2) **Angular Velocity:** Using the unwrapped version of the angle yields a continuous function suitable for differentiation. However using the timeseries which have already been de-noised and quadratically smoothened will still be wrapped. Apply differentiation will amplify high-frequency noise, so apply Savitzky-Golar filter

$$\varphi_i = \text{SG}(\varphi_i, W, p, \text{deriv} = 0), \quad (3)$$

$$\omega_i = \dot{\varphi}_i = \text{SG}(\varphi_i, W, p, \text{deriv} = 1, \delta = \Delta t). \quad (4)$$

Optionally $\alpha_i = \ddot{\varphi}_i$ from the same filter ('deriv=2'); it is not needed for U , V , L but can be used in the consistency checks

3) **Magnets Positions:** In the sticks body frame exist a magnet, at (r, a) on the long rod and $(e, -b)$ on the short stick

$$\mathbf{p}_i(t) = O + R(\varphi_i^s(t)) \mathbf{r}_{\text{mag},i}, \quad R(\varphi) = \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix}. \quad (5)$$

A. Potential Energy $U(t)$

$$U(t) = \underbrace{g [m_L d_L \cos \beta_L(t) + m_S d_S \cos \beta_S(t)]}_{V_{\text{grav}}(t)} + \underbrace{U_{\text{mag}}(\rho(t), \hat{\mathbf{s}}(t), \varphi_L, \varphi_S)}_{V_{\text{mag}}(t)} + C. \quad (6)$$

- V_{grav} also includes the magnet point masses. May add

$$g m_{\text{mag},i} \rho_{\text{mag},i} \cos \beta_{\text{mag},i}(t)$$

per magnet, with the same form but a different lever arm and phase.

- Magnetic term, point-dipole model(not certain whether we should use include this effect of the magnet:

$$U_{\text{mag}} = \frac{\mu_0}{4\pi \rho^3} [\mu_1 \cdot \mu_2 - 3(\mu_1 \cdot \hat{\mathbf{s}})(\mu_2 \cdot \hat{\mathbf{s}})]. \quad (7)$$

Here, each μ_k is rotated into the frame by $R(\varphi_k)$.

- The constant C is chosen such that

$$\min_t U(t) = 0$$

over the analyzed record.

B. Kinetic Energy $V(t)$

$$V(t) = \frac{1}{2} I_L \omega_L(t)^2 + \frac{1}{2} I_S \omega_S(t)^2 \quad (8)$$

We determine the total and the per-stick split V_L, V_S . V may be dominated by differentiation error

C. Total angular momentum $L(t)$

$$L(t) = I_L \omega_L(t) + I_S \omega_S(t) \quad (9)$$

Report the total and the per-stick split. Note that L about O is not conserved: gravity exerts torque $\tau_{\text{grav}}(t) = -g[m_L d_L \sin \beta_L + m_S d_S \sin \beta_S]$ about O , and the magnetic coupling exchanges angular momentum between the rods and, to a small extent, with the field. The measured $L(t)$ and its derivative are therefore also diagnostics

D. Total Mechanical Energy

$$E(t) = U(t) + V(t) \quad (10)$$

$E(t)$ is produced alongside the timeseries and using this we determine the energy contribution from the short and long stick.

V. EMBEDDING AND FEATURE CONSTRUCTION

In building set if supervised input vector and output forecast labels for for over the top or next position prediction, the choice of which feature is very important for characterize the best state or coordinate representation that is able to capture the dynamics of the kinetic swinging sticks motion. For forecasting, generally constructing the X via Takens' delay embedding theorem with embedding dimension and delay determined empirically depending of the system being studied. However, the choice of input use for the delay embedding vectors affects the accuracy of your forecast since they form or used as basis for determining the mapping of a state to it prediction next state.

Fundamentally, obtaining a good model is linked to the coordinate system in which the dynamics are measured. Considering the state of the long or short stick is represented as an angle, forecasting the next state which is also an angle in degrees or radian, using Takens' delay embedding of timeseries is simple just as forecasting a time to the top, will required Taken embedding of over the top time positions to for X used in building the mapping for each forecast.

However, multiple information in data presents different ways of building a training set data X_L and Y_L ; using time, angle, angular velocity, time to over the top or change in time to over the top and clock hour positions.

For whichever information set used in building X_L and Y_L , a model builds a map of using predicting target y_t from x_t . For prediction using local linear model at the core of method that leverages a kernel for determined weights used in

A. Computing Distance in Delay Vectors of Angles

The first principles of a local linear model is selection of vectors that represent or characterize the

B. Circular Delay Embedding for the Angular State Evolution

The stick angle θ recorded by the tracking algorithm of in the appendix is not a general scalar; not just a monotonously increasing or decreasing value: it is an element of the circle $\theta \in [0^\circ, 360^\circ)$ with the identification $0^\circ \equiv 360^\circ$, i.e. a point of $\mathbb{R}/360^\circ, S^1$. The delay-embedding construction of from Takens' Section builds, at every time index i , a vector of d lagged observations separated by delay τ ,

$$\mathbf{v}_i = [\theta_i, \theta_{i-\tau}, \theta_{i-2\tau}, \dots, \theta_{i-(d-1)\tau}], \quad (11)$$

which the local-linear predictor (Section V-B3) uses both to search for analogue states and as the model for $\hat{x}_{t+1} \approx \mathbf{A}\mathbf{x}_t + \mathbf{b}$. At the core or fundamental stage of this algorithm hinges on one further goal: a notion of distance between two delay vectors, used to select the k nearest neighbours the local-linear fit is built from. We first show that the distance used in our old embedding representation is the wrong metric for studying or predicting the dynamics of a periodic kinetic swinging system, and then give the embedding under which it becomes the correct one.

1) *Baseline: Euclidean Distance on Raw-Degree Vectors:* Nearest-neighbour search over the training set $\{\mathbf{v}_j\}_{j=1}^{N_{\text{train}}}$ is performed with a kD-tree, which indexes points in $\mathbb{R}^d$ and answers a query $\mathbf{v}_i$ by returning the k training vectors minimising the Minkowski- p distance, with $p = 2$ (2-norm) as the default. Time indices are excluded from the indexed coordinates, so the KD-tree operates purely on the d lagged angle values, and the distance between two raw delay vectors $\mathbf{v}_i, \mathbf{v}_j \in \mathbb{R}^d$ is

$$d_{\text{raw}}(\mathbf{v}_i, \mathbf{v}_j) = \left(\sum_{l=0}^{d-1} (\theta_i^{(l)} - \theta_j^{(l)})^2 \right)^{1/2}. \quad (12)$$

Equation (12) is the correct metric on $\mathbb{R}^d$, but the coordinates $\theta^{(l)}$ do not live in $\mathbb{R}^d$: each lies on S^1 , and (12) silently treats S^1 as if it were cut open into the interval $[0^\circ, 360^\circ)$ and laid flat on the real line. That cut introduces a discontinuity exactly where none exists physically. Take a query angle $\theta_i = 349^\circ$ as the last current state in delay vector against two library candidates delay vectors with last vector values $\theta_A = 5^\circ$ and $\theta_B = 200^\circ$. The true angular separations, measured the short way around the circle, are 16° (θ_i to θ_A) and 149° (θ_i to θ_B) – A is by far the closer point. Equation (12), evaluated on the open interval, gives instead $|349^\circ - 5^\circ| = 344^\circ$ and $|349^\circ - 200^\circ| = 149^\circ$: the KD-tree ranks B as the nearer neighbour, the opposite of the physical ordering. Every delay window whose lags straddle the $0^\circ/360^\circ$ boundary is subject to this misranking, which is not a rare edge case for a system whose long stick completes full revolutions or over the top crossings.

The error does not stop at neighbour selection. Even a distance metric that correctly ranked neighbours near the wrap (e.g. a periodickD tree) would leave the regression step

in error, because the local-linear fit $\mathbf{A}\mathbf{x}_t + \mathbf{b}$ is estimated by ordinary least squares over the k neighbour targets. A neighbourhood of angles genuinely clustered near the wrap, e.g. $\{358^\circ, 2^\circ, 359^\circ, 1^\circ\}$, is clustered near 0° physically, but its arithmetic mean under (11)'s flat-interval representation is $\approx 180^\circ$, utterly wrong. Ordinary least squares computes exactly such an arithmetic average implicitly; the raw-degree representation therefore corrupts the target space representation of the regression independently of how the neighbors were found. This second effect is the reason the fix below changes the representation itself rather than only the distance function used for the kD tree search.

2) *Improved Representation: Embedding onto the Unit Circle:* The standard remedy in directional statistics for placing an angular variable into a Euclidean space without introducing a cut point is the embedding

$$\Phi: \theta \mapsto (\cos \theta, \sin \theta) \in S^1 \subset \mathbb{R}^2, \quad (13)$$

which is continuous and injective on $[0^\circ, 360^\circ)$, so no two physically distinct angles are ever identified, and no boundary is introduced where none exists. Applying Φ to every lag of (11) in place of the raw angle gives the *circular delay vector*

$$\tilde{\mathbf{v}}_i = [\cos \theta_i^{(l)}, \sin \theta_i^{(l)}]_{l=0}^{d-1} \in \mathbb{R}^{2d}, \quad \theta_i^{(l)} := \theta_{i-l\tau}, \quad (14)$$

i.e. every lag doubles from one raw-degree coordinate to a $(\cos, \sin)$ pair, and the kD tree is built over this $2d$ -dimensional space instead, *the search algorithm itself is unchanged*; only the geometry of the space it searches is corrected.

The Euclidean distance between two points on the image of Φ reduces to a closed form. For two angles θ_i, θ_j ,

$$\begin{aligned} \|\Phi(\theta_i) - \Phi(\theta_j)\|_2^2 &= (\cos \theta_i - \cos \theta_j)^2 + (\sin \theta_i - \sin \theta_j)^2 \\ &= 2 - 2 \cos(\theta_i - \theta_j) \\ &= 4 \sin^2\left(\frac{\theta_i - \theta_j}{2}\right), \end{aligned} \quad (15)$$

using $\cos(a - b) = \cos a \cos b + \sin a \sin b$ and the half-angle identity $1 - \cos x = 2 \sin^2(x/2)$. Hence

$$d_{\text{chord}}(\theta_i, \theta_j) := \|\Phi(\theta_i) - \Phi(\theta_j)\|_2 = 2 \left| \sin\left(\frac{\theta_i - \theta_j}{2}\right) \right|, \quad (16)$$

the *chord distance* on the unit circle. Unlike (12), d_{chord} is a strictly increasing function of the true angular separation $\Delta\theta = \min(|\theta_i - \theta_j|, 360^\circ - |\theta_i - \theta_j|) \in [0^\circ, 180^\circ]$, so ranking candidates by (16) is equivalent to ranking them by physical closeness on the circle, with no discontinuity at $0^\circ/360^\circ$: d_{chord} is bounded in $[0, 2]$ and attains its maximum only at the true antipode $\Delta\theta = 180^\circ$. Summed over the d lags, the distance between two circular delay vectors is

$$d_{\text{circ}}(\tilde{\mathbf{v}}_i, \tilde{\mathbf{v}}_j) = \left(\sum_{l=0}^{d-1} 4 \sin^2\left(\frac{\theta_i^{(l)} - \theta_j^{(l)}}{2}\right) \right)^{1/2}, \quad (17)$$

which is again an ordinary ℓ_2 norm – now correctly applied, since it is computed in the ambient Euclidean space $\mathbb{R}^{2d}$ that

$S^1 \times \dots \times S^1$ is embedded in, rather than in the periodic space itself.

Re-evaluating the worked example of Section V-B1 under (16): $d_{\text{chord}}(349^\circ, 5^\circ) = 2 \sin(8^\circ) = 0.28$ and $d_{\text{chord}}(349^\circ, 200^\circ) = 2 \sin(74.5^\circ) = 1.93$. ThekD tree now correctly ranks A as the nearer neighbour ($0.28 < 1.93$), matching the true 16° versus 149° angular separations.

3) *Regression and Forecast Recovery*: With neighbours selected under (17), the local-linear model is fit jointly on the two-dimensional target $\mathbf{y} = (\cos \theta, \sin \theta)$ rather than on the raw angle,

$$\hat{\mathbf{y}}_{t+1} \approx \mathbf{A} \mathbf{x}_t + \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{2 \times 2d}, \mathbf{b} \in \mathbb{R}^2, \quad (18)$$

by ridge-regularised least squares over the same k neighbours, and the forecast angle is recovered by

$$\hat{\theta}_{t+1} = \text{atan2}(\hat{y}_{t+1}, \hat{x}_{t+1}) \bmod 360^\circ. \quad (19)$$

Because atan2 is defined for every $(\hat{x}_{t+1}, \hat{y}_{t+1}) \neq (0, 0)$, (19) always returns a valid angle in $[0^\circ, 360^\circ)$ with no clamping of the model output – under the raw-degree representation of Section V-B1, by contrast, the unconstrained regression output can and does fall outside $[0^\circ, 360^\circ)$ whenever a neighbourhood spans the wrap, which is the direct mechanism behind the largest observed errors under that baseline (Section V-B4). The residual magnitude $r = \sqrt{\hat{x}_{t+1}^2 + \hat{y}_{t+1}^2}$ is a byproduct of (18) with no counterpart in the raw-degree pipeline: $r \approx 1$ indicates the k neighbours agreed closely on a forecast direction, while $r \rightarrow 0$ flags a neighbourhood whose targets pointed in conflicting directions, for which $\hat{\theta}_{t+1}$ is correspondingly ill-conditioned. Forecast error itself is scored with the matching wrap-aware quantity,

$$e = ((\hat{\theta}_{t+1} - \theta_{t+1}) + 180^\circ) \bmod 360^\circ - 180^\circ, \quad (20)$$

bounded to $(-180^\circ, 180^\circ]$, so that a 359° prediction against a 1° target scores as the physically correct 2° error rather than 358° .

4) *Empirical Comparison*: Table II compares the two embeddings on the long-stick series (60 fps source, decimated to $\Delta t = 0.5$ s, $d = 5$, $\tau = 3$ steps (1.5 s), forecast horizon 3 steps (1.5 s ahead), $k = 16$), holding the local-linear predictor, the delay parameters, and the temporal train/test split identical across rows. Error is reported under the common circular metric (20) in both rows, so that the raw-degree embedding’s own error column is not inflated relative to the circular embedding’s by definition. The circular embedding at $N_{\text{train}} = 1024$ outperforms the raw-degree embedding even at $N_{\text{train}} = 4096$, indicating that at this lead time the state-space representation, not the size of the training library, is the binding constraint on forecast skill.

Beyond the aggregate error reduction, the raw-degree embedding’s largest errors are structurally different in kind from the circular embedding’s: because the raw-degree regression output is unconstrained, its worst-case errors exceed 360° in magnitude on this dataset, a signature of the mechanism

TABLE II
LONG-STICK FORECAST ERROR BY DELAY-VECTOR REPRESENTATION, EVALUATED UNDER THE COMMON WRAP-AWARE ERROR (20). $d = 5$, $\tau = 3$ STEPS (1.5 S), HORIZON = 3 STEPS (1.5 S AHEAD), $k = 16$, $N_{\text{test}} = 2000$.

Embedding	N_{train}	RMSE ($^\circ$)	MAE ($^\circ$)
Raw-degree, $\mathbb{R}^d$	1024	83.9	66.5
Raw-degree, $\mathbb{R}^d$	4096	72.3	53.3
Circular, $\mathbb{R}^{2d}$	1024	65.1	46.0

described in Section V-B3 rather than of ordinary forecast difficulty. The circular embedding’s error is bounded by construction to 180° regardless of how poorly a given neighbourhood agrees, since (19) cannot produce an angle outside $[0^\circ, 360^\circ)$. The remaining error at 65° – 84° RMSE is not attributable to the representation: it is close to uniform across the swing (not concentrated at the sharp reversals near the top and bottom of each oscillation, nor correlated with local angular speed), consistent instead with a forecast horizon that is an appreciable fraction of the system’s natural swing period, and with the embedding being univariate in the long-stick angle alone despite the coupled, two-arm nature of the physical system.

VI. RESULTS

In this, we provide results obtained from baseline ablation study from Echo State Networks (ESN), Support Vector Regression (SVR), Gaussian Process Regression or Bayesian (Probabilistic) (GPR), Multilayer Perceptron (MLP), Random Forest Regression (RF), Koopman operator (EDMD) and Long Short Term Memory (LSTM). For simpler prediction, we used the delay embedded vector angle representation.

A. Ablation Protocol

For each model, we held all hyper-parameters at their default values and varied one hyperparameter at a time (*one-at-a-time* or OAT ablation). This isolates the contribution of each parameter to the overall predictive performance. A full grid search is computationally intractable for all seven models jointly; OAT ablation provides directional sensitivity and is standard in reservoir computing literature [12].

B. Models Evaluation Table for Long and Short Sticks Ablation

Table ?? reports the test-set performance of each model with its best configuration identified during the ablation study. The GPR achieves the lowest NRMSE, consistent with its theoretical advantage as a probabilistic model with inherent memory matching the temporal structure of chaotic maps. GPR is competitive with ESN but is limited by its $\mathcal{O}(N^3)$ complexity (trained on $N = 800$ samples here). SVR with RBF kernel performs well and offers strong practical guarantees via its convex optimization. MLP and RF are the most computationally efficient models and achieve competitive but slightly inferior NRMSE.

VII. CONCLUSION

We introduced an experimental dataset from the long-stick angle of the Swinging Sticks kinetic sculpture and performed a broad comparison of seven nonlinear time-series forecasters. Our ablation study on long sticks data, by systematically varying each model's key hyperparameters, showed that Bayesian kernel methods (GPR, NRMSE = 0.00112) and operator-theoretic approaches (EDMD, NRMSE = 0.00620) achieve the lowest normalized errors on one-step-ahead prediction. Simpler ensemble and feedforward models (RF, MLP) were nearly as accurate and substantially cheaper to train, while deep recurrent architectures (LSTM, NRMSE = 0.15140) were over-parameterised relative to the available data, incurring training times of 65–145 s with no commensurate accuracy gain.

For our short stick timeseries forecast reveals an important shift in model ranking. SVR (NRMSE = 0.02246, R^2 = 0.99950), displaces GPR at the top, as the increased trajectory irregularity degrades the smoothness assumption underlying GPR's kernel. RF and EDMD remain competitive, while LSTM collapses (R^2 = 0.65), reinforcing that deep recurrent architectures are data-hungry relative to the available training set. Across both arms of the Swinging Sticks, kernel and ensemble methods prove the most reliable forecasters.

In the future, we plan to leverage these findings in the system identification process as we attempt to infer interpretable governing equations of the Swinging Sticks. Since the underlying physics can be obtained analytically, data-driven methods such as SINDy or Koopman eigenfunctions may extract its dynamical equations directly from data. This work lays the groundwork by identifying the most promising learning models for capturing the chaotic dynamics of the Swinging Sticks system.

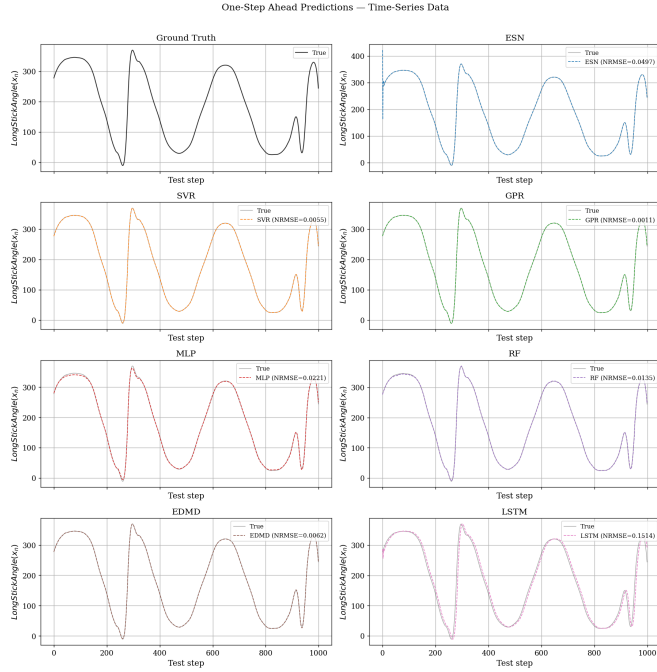
ACKNOWLEDGMENT

VIII. DATASET AND CODE

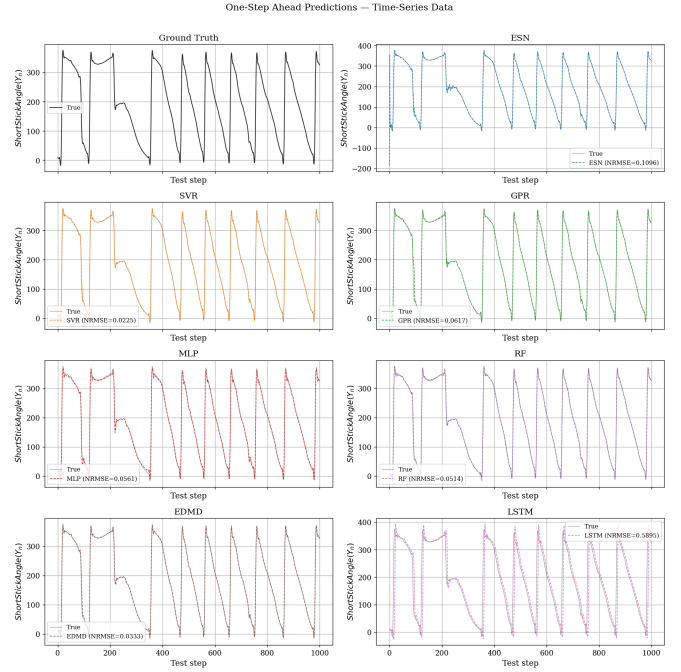
The dataset and code used in this work are not publicly released during the review process to preserve the double-blind review policy and due to other restrictions. The authors will make the code and dataset available upon reasonable request and plan to release them publicly after acceptance.

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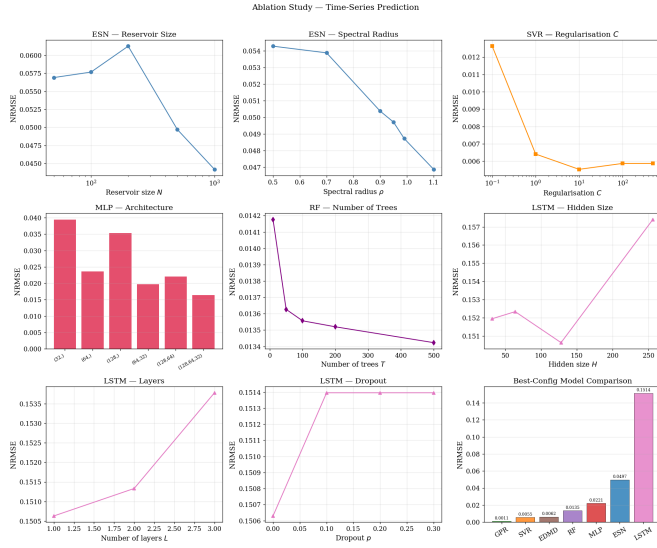
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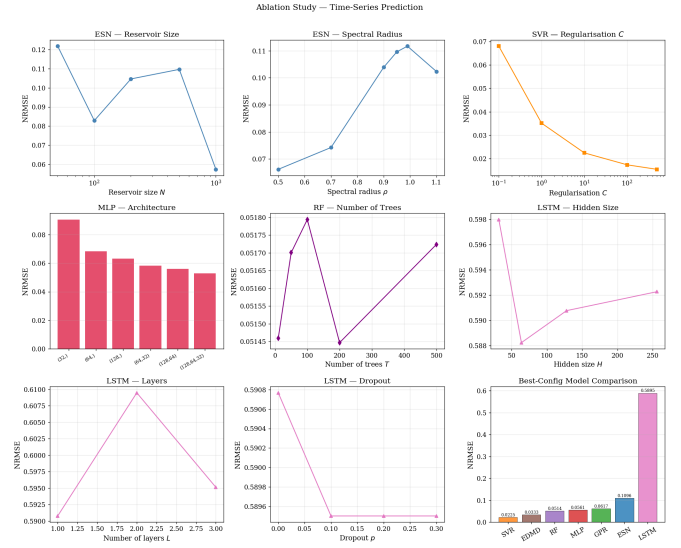
(a) Long Stick Predictions



(b) Short Stick Predictions



(c) Long Stick Ablation Metrics



(d) Short Stick Ablation Metrics

Fig. 9. Top row: Comparison of one-step-ahead predictions for the long-stick angle and short-stick angle time series. The ground-truth trajectory is shown alongside model predictions for ESN, SVR, GPR, MLP, RF, EDMD, and LSTM. The NRMSE metric quantifies predictive accuracy. Kernel-based methods (GPR, SVR) and operator-theoretic EDMD demonstrate superior accuracy with minimal phase lag and reduced error near sharp nonlinear transitions, whereas recurrent (LSTM) and reservoir-based (ESN) models show larger deviations in highly nonlinear regions. Bottom row: Ablation study metrics for long- and short-stick time series.